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Subj

C R RAO Advanced Institute of Mathematics, Statistics and Computer Science (AIMSCS) in Collaboration with Jawaharlal Nehru Technological University Hyderabad

3rd year B.Tech CSE DS 2nd Semester 2nd MID Examinations, May- 2025

Subject: Cryptography and Network Security **DoE:22-05-2025**

Time: 120 Minutes **Max. Marks: 30 M**

PART-A

Answer all questions **5X2=10M**

1. What is Message Authentication Code (MAC).
2. Briefly explain the Elgamal Digital Signature Scheme.
3. What is HTTPs and how it works.
4. List out Email based Attacks.
5. Describe Authentication Header.

PART – B

Answer any FOUR questions **4X5 = 20 M**

1. Explain X.509 authentication and certificates.
2. Describe Kerberos in detail.
3. Explain SSL (Secure Socket Layer) Architecture in detail.
4. Explain SSH Protocol Stack.
5. What is Pretty Good Privacy. Explain its operations.
6. Explain Shamir's (t,n) Secret sharing scheme.

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3. (a) What is hash function. Explain properties of Hash function in detail.
(b) Describe SHA-512 algorithm.

OR

(c) Explain X.509 authentication and certificates.
(d) Explain ECC Digital Signature Scheme.

4. (a) Explain SSL (Secure Socket Layer) Architecture in detail.
(b) Compare WEP, WPA, and WPA2 in terms of encryption strength and vulnerabilities.

OR

(c) Explain SSH Protocol Stack.
(d) Explain Pretty Good Privacy and its operations.

5. (a) Explain E-Mail Security. What are the Email Security Enhancements.
(b) Explain IP Sec &IP Security architecture.

OR

(c) Describe Encapsulating security payload.
(d). Explain Shamir's (t,n) Secret sharing scheme.

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R22

Subject Code: C3204

C R RAO Advanced Institute of Mathematics, Statistics and Computer Science (AIMSCS) in Collaboration with Jawaharlal Nehru Technological University Hyderabad

3rd year B.Tech 2nd Semester Final Examinations, June-2025

Branch: B.Tech CSE - Data Science

Subject: Cryptography and Network Security DoE:09-06-2025

Time: 180 Minutes Max. Marks: 60 M

PART-A 10X1=10M

Answer all questions

1. What are the Security Goals.
2. What is the fundamental difference between substitution and transposition techniques?
3. Difference between Symmetric key and Asymmetric key cryptography.
4. Explain Message authentication codes.
5. Write difference between Digital signature and Digital certificate.
6. Explain Public – Key Infrastructure.
7. What are the differences between WPA and WPA2?
8. What are the Web security considerations
9. What is S/MIME.
10. Explain PGP Message Format.

PART – B

Answer all questions 5X10 = 50 M

1. (a) Explain Security attacks, Security services, Security Mechanisms.
   (b) Explain in detail model for Network Security Cryptography

   OR

   (c) Describe the Vigenère cipher and provide an encryption example.
   (d) Differentiate between Autokey, and Running Key ciphers, with examples.

2. (a) In RSA, if the public key is (e = 13, n = 187) and the ciphertext is $C = 11$, find the plaintext message.
   (b) Explain AES Cryptosystem Encryption algorithm.

   OR

   (c) Explain modes of operations.
   (d) Explain Diffe-Hellman algorithm with example.